

Review article: Numerical Methods for Engineering Quantum Computers

Liu, Shu

Mathematics, Florida State University

Georgiadou, Antigoni

Oak Ridge National Laboratory

Dong, Spencer

Oak Ridge National Laboratory

Kubischta, Eric

Mathematics, Florida State University

Melton, Daniel

Mathematics, Florida State University

Guenther, Stefanie

Lawrence Livermore National Laboratory

Jin, Shi

Mathematics, Shanghai Jiao Tong University

Lin, Lin

Mathematics, U.C. Berkeley

Lotshaw, Philip

Oak Ridge National Laboratory

Petersson, Anders

Lawrence Livermore National Laboratory

Sussman, Mark

Mathematics, Florida State University

October 22, 2025

ABSTRACT: In the article by Herrmann et al, “Quantum utility – definition and assessment of a practical quantum advantage,” the “Application Readiness Level” (ARL) is defined: (ARL1) “concept with unknown potential,” (ARL2) “beneficial in small idealized systems,” (ARL3) “utility indicated by

theory or resource estimations,” (ARL4) “simulated utility demonstration,” and (ARL5) “utility demonstration on quantum hardware.” At the present, practitioners have reached ARL3 with the Variational Quantum Eigenvalue (VQE) algorithm. This seems to indicate that there is still much work to be done in order to reach ARL4 or ARL5. In this review article, the latest activities from the computational “digital twin” perspective are reported on in which scientists are working hard to bring practitioners out of the Noisy Intermediate Scale Quantum (NISQ) era into an era of “Quantum supremacy.” In other words this review article discusses digital twin activities that help to advance from ARL3 to ARL4, or even ARL5. The three main categories in this review article are (1) Quantum algorithms for benchmarking quantum computers (to include emulator algorithms like Qiskit), (2) Optimization, Control, inverse problems, and exploratory simulation algorithms for Schrodinger’s equation, and (3) Quantum Error correction algorithms and surface codes.

1 Introduction

The motivation for this review article is the fact that Moore’s law is no longer applicable[77]. As quoted from Shalf[90], “Moore’s Law [1] is a techno-economic model that has enabled the IT industry to double the performance and functionality of digital electronics roughly every 2 years within a fixed cost, power and area. This expectation has led to a relatively stable ecosystem (e.g. electronic design automation tools, compilers, simulators and emulators) built around general-purpose processor technologies, such as the $\times 86$, ARM and Power instruction set architectures. However, within a decade, the technological underpinnings for the process that Gordon Moore described will come to an end, as lithography gets down to atomic scale. At that point, it will be feasible to create lithographically produced devices with dimensions nearing atomic scale, where a dozen or fewer silicon atoms are present across critical device features, and will therefore represent a practical limit for implementing logic gates for digital computing [2]. Indeed, the ITRS (International Technology Roadmap for Semiconductors), which has tracked the historical improvements over the past 30 years, has projected no improvements beyond 2021, as shown in figure 1, and subsequently disbanded, having no further purpose. The classical technological driver that has underpinned Moore’s Law for the past 50 years is failing [3] and is anticipated to flatten by 2025, as shown in figure 2. Evolving technology in the absence of Moore’s Law will require an investment now in computer architecture and the basic sciences (including materials science), to study candidate replacement materials and alternative device physics to foster continued technology scaling.”

References one through three in the above quote correspond to the following references respectively: [77][74][75].

As further motivation for this review article, in Figure 3 of the article by Shalf[90], a roadmap is provided for possible paths forward when the density of circuits on classical computers exceeds a critical value. There are three cat-

egories: (a) roadmap for the next ten years, (b) 20 years, and (c) “Decades beyond exascale,” “New Models of Computation.” For category (c), Shalf[90] refers to the following prospective technology: (i) approximate computing, (ii) adiabatic reversible, (iii) Analog, (iv) Neuromorphic, and (v) quantum[27, 28]. Of the “new models of computation” listed in Shalf’s article[90], this review article will address the “Numerical Methods for Engineering Quantum Computers” aspects associated with the emerging “quantum computing” paradigm. As outlined by Gamble[29], the development of reliable quantum computers will have a transformative effect on computer technology. That being said, right now we are in the “Noisy Intermediate-Scale Quantum”[14] (NISQ) era. The purpose of this review article is to bring to the forefront the current research being done, from the computational perspective, in order to move beyond the NISQ era.

2 Quantum algorithms and Quantum emulators for benchmarking quantum computers

Quantum Emulators:

1. (IBM) “Quantum Computing with Qiskit” (2024)[48]
2. “Intel Quantum Simulator: a cloud-ready high-performance simulator of quantum circuits” (2020)[34]
3. (Microsoft) “LIQUid: A Software design architecture and domain specific language for quantum computing” (2014)[100]
4. “QuESTlink—Mathematica embiggened by a hardware-optimised quantum emulator” (2020)[52]
5. (Matlab) “QCLAB: A Matlab Toolbox for Quantum Computing” (2025)[55]
6. “PennyLane-Lightning MPI: A massively scalable quantum circuit simulator based on distributed computing in CPU clusters” (2025)[54]
7. (Google) “qsim”(2020)[97]
8. (nVidia) “cuQuantum SDK: A High-Performance Library for Accelerating Quantum Science” (2023)[7]
9. “Numerical recipes in quantum information theory and quantum computing: an adventure in FORTRAN 90”(2021)[85]
10. Nguyen et al(2022)[80] “Tensor network quantum virtual machine for simulating quantum circuits at exascale”
11. Zhang et al (2024)[104], “Quantum circuit simulation with fast tensor decision diagram”

12. Viamontes et al (2004)[99], “High-performance QuIDD-based simulation of quantum circuits.”

Quantum Algorithms for benchmarking Quantum Computers:

1. Kitaev (1995)[59], “Quantum measurements and the Abelian stabilizer problem” (quantum phase estimation)
2. Shor (1999)[93], “Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer”
3. Callison and Chancellor (2022)[14] “Hybrid quantum-classical algorithms in the noisy intermediate-scale quantum era and beyond”
4. Jin et al (2024)[49], “Quantum Simulation for Quantum Dynamics with Artificial Boundary Conditions.”
5. S. Jin, N. Liu, and Y. Yu (2024)[51], “Quantum Circuits for the heat equation with physical boundary conditions via Schrodingerisation”
6. Cerezo et al (2021)[15] “Variational quantum algorithms”
7. Lotshaw et al (2021)[71] “Simulations of frustrated Ising Hamiltonians using quantum approximate optimization”
8. Nielsen and Chuang (2010)[81] “Quantum computation and quantum information” (quantum Fourier Transform and quantum phase estimation)
9. Ding and Lin (2023)[21] “Simultaneous estimation of multiple eigenvalues with short-depth quantum circuit on early fault-tolerant quantum computers”
10. Biamonte et al (2017)[9] “Quantum machine learning”
11. Benedetti et al (2019)[8] “Parameterized quantum circuits as machine learning models”
12. Khait et al(2023)[56] ”Variational quantum eigensolvers in the era of distributed quantum computers”
13. Sawaya et al (2023)[88], “HamLib: A Library of Hamiltonians for Benchmarking Quantum Algorithms and Hardware”
14. Georgadou et al(2024)[32] Hele-Shaw simulation on quantum computer.
15. Kim et al (2023)[58], “Evidence for the utility of quantum computing before fault tolerance”
16. Zoufal et al (2019)[107], “Quantum generative adversarial networks for learning and loading random distributions”
17. Nghiem and Tzu-Chieh (2023)[79], “Quantum algorithm for estimating largest eigenvalues”

18. Cadi et al (2024)[13], “Folded spectrum vqe: A quantum computing method for the calculation of molecular excited states”
19. Tilly et al (2022)[98], “The variational quantum eigensolver: a review of methods and best practices”
20. Qing and Xie (2023)[84], “Use VQE to calculate the ground energy of hydrogen molecules on IBM Quantum”
21. Ryu et al (2021)[87], “A Variational Quantum Algorithm for Ordered SVD”
22. Shirakawa et al (2024)[91], “Automatic quantum circuit encoding of a given arbitrary quantum state”
23. Harrow et al (2009)[40], “Quantum algorithm for linear systems of equations”
24. Daribayev et al (2023)[18], “Implementation of the hhl algorithm for solving the poisson equation on quantum simulators.”
25. Bravo-Prieto et al (2023)[11], ”Variational quantum linear solver”
26. Gilyen et al (2019)[30], “Quantum singular value transformation and beyond: exponential improvements for quantum matrix arithmetics”
27. Gundlach et al (2025)[35], “Quantum Advantage in Computational Chemistry?”
28. Ko et al (2023)[62], “Implementation of the density-functional theory on quantum computers with linear scaling with respect to the number of atoms”
29. Jin et al (2025)[50], “Quantum preconditioning method for linear systems problems via Schrodingerization”
30. Itani et al (2024)[47], “Quantum algorithm for lattice Boltzmann (QALB) simulation of incompressible fluids with a nonlinear collision term”
31. Hibat-Allah et al (2024)[45], “A framework for demonstrating practical quantum advantage: comparing quantum against classical generative models”
32. Herrmann et al (2023)[44], “Quantum utility–definition and assessment of a practical quantum advantage”
33. Lotshaw et al (2023)[69], “Modeling noise in global Molmer-Sorensen interactions applied to quantum approximate optimization.”
34. Di Bartolomeo et al (2023)[20], “Noisy gates for simulating quantum computers.”

35. De Luo et al (2025)[72], “Quantum simulation of bubble nucleation across a quantum phase transition.”
36. Maciejewski et al (2024)[73], “Improving Quantum Approximate Optimization by Noise-Directed Adaptive Remapping.”

3 Optimization, Control, inverse problems, and exploratory simulation algorithms for Schrodinger’s equation

1. Feit et al (1982)[26], “Solution of the Schrödinger equation by a spectral method.”
2. Bao et al (2002)[4], “On Time-Splitting Spectral Approximations for the Schrödinger Equation in the Semiclassical Regime.”
3. Lin et al (2016)[68], “Approximating spectral densities of large matrices.”
4. Zhang et al (2017)[103], “Observation of a many-body dynamical phase transition with a 53-qubit quantum simulator.”
5. Han et al (2019)[38], “Solving many-electron Schrödinger equation using deep neural networks.”
6. Günther et al (2021) [37], “Quandary: An open-source C++ package for high-performance optimal control of open quantum systems.” the article by Lidar (2019)[66] (“Lecture notes on the theory of open quantum systems”) gives the derivation of the model for open quantum systems.
7. Günther and Petersson (2023) [36], “A practical approach to determine minimal quantum gate durations using amplitude-bounded quantum controls.”
8. Fei et al (2025)[24], “Switching time optimization for binary quantum optimal control.”
9. Fei et al (2023)[25], “Binary control pulse optimization for quantum systems.”
10. Kanai et al (2024)[53], “Single-Electron Qubits Based on Quantum Ring States on Solid Neon Surface.”
11. Hermann et al (2020)[43], “Deep-neural-network solution of the electronic Schrödinger equation.”
12. Andrade et al (2022)[2], “Engineering an effective three-spin Hamiltonian in trapped-ion systems for applications in quantum simulation.”

13. Wille et al (2018)[101], “Computer-aided design for quantum computation.”
14. He et al (2024)[42], “Efficient Optimal Control of Open Quantum Systems”
15. Nusseler et al (2020) [82], “Efficient simulation of open quantum systems coupled to a fermionic bath.”
16. Selsto and Kvaal (2010)[89], “Absorbing boundary conditions for dynamical many-body quantum systems.”
17. Sawaya et al (2023)[88], “HamLib: A Library of Hamiltonians for Benchmarking Quantum Algorithms and Hardware.”
18. Grivopoulos (2005)[33], “Optimal control of quantum systems.”
19. Hangleiter et al (2024)[39], “Robustly learning the Hamiltonian dynamics of a superconducting quantum processor.”
20. Shukrinov et al (2016)[94], “Modeling of LC-shunted intrinsic Josephson junctions in high-Tc superconductors.”
21. Sun and Galperin (2025)[96], “Control of open quantum systems: The nonequilibrium Green’s function perspective.”
22. Zhang et al (2004)[105], “Numerical solutions of the Schrodinger equation for the ground lithium by the finite element method.”
23. He and Wu (2024)[41], “Zeroing neural dynamics solving time-variant complex conjugate matrix equation”
24. Liao et al (2020)[65], “Modified gradient neural networks for solving the time-varying Sylvester equation with adaptive coefficients and elimination of matrix inversion”
25. Lin Lin (2025)[67], “Dissipative preparation of many-body quantum states: Toward practical quantum advantage”

4 Quantum Error Correction algorithms and surface codes

1. Shor (1995)[92], “Scheme for reducing decoherence in quantum computer memory.”
2. Anthony-Petersen et al (2024)[3], “A stress-induced source of phonon bursts and quasiparticle poisoning.”
3. Lotshaw et al (2024)[70], “Exactly solvable model of light-scattering errors in quantum simulations with metastable trapped-ion qubits.”

4. Kubischta, Eric and Teixeira, Ian (2023)[64], “Family of quantum codes with exotic transversal gates.”
5. Barends et al (2014)[5], “Superconducting quantum circuits at the surface code threshold for fault tolerance.”
6. Baum et al (2021)[6], “Experimental deep reinforcement learning for error-robust gate-set design on a superconducting quantum computer.”
7. Cheng and Guo (2025)[16], “Modeling correlated-noise in silicon spin qubit device.”
8. Putterman et al (2025)[83], “Hardware-efficient quantum error correction via concatenated bosonic qubits”
9. Knill and Laflamme (1997)[61], “Theory of quantum error-correcting codes”
10. Eastin and Knill (2009)[22], “Restrictions on transversal encoded quantum gate sets”
11. Ide et al (2025)[46], “Fault-tolerant logical measurements via homological measurement”
12. Bravyi and Kitaev (2002)[12], “Fermionic quantum computation”
13. Dennis et al (2002)[19], “Topological quantum memory”
14. Kitaev (2003)[60], “Fault-tolerant quantum computation by anyons”
15. Roffe (2019)[86], “Quantum error correction: an introductory guide”
16. Bluvstein et al (2024)[10], “Logical quantum processor based on reconfigurable atom arrays”
17. Marra (2022)[76], “Majorana nanowires for topological quantum computation”
18. Evered et al (2023)[23], “High-fidelity parallel entangling gates on a neutral-atom quantum computer”
19. Wintersperger et al (2023)[102], “Neutral atom quantum computing hardware: performance and end-user perspective”
20. Gonthier et al (2022)[31], “Measurements as a roadblock to near-term practical quantum advantage in chemistry: Resource analysis”
21. Dalton et al (2024)[17], “Quantifying the effect of gate errors on variational quantum eigensolvers for quantum chemistry”
22. Nautrup et al (2019)[78], “Optimizing quantum error correction codes with reinforcement learning”

23. Krinner et al (2022)[63], “Realizing repeated quantum error correction in a distance-three surface code”
24. Stetina and Wiebe (2025)[95], “First-Quantized Quantum Simulation of Non-Relativistic QED with Emergent Topologically Protected Coulomb Interactions”
25. Zou and Hoi-Kwong (2025)[106], “Algebraic Topology Principles behind Topological Quantum Error Correction”
26. Google Quantum (2025)[1], “Quantum error correction below the surface code threshold”
27. Khan et al (2025)[57], “Separate and efficient characterization of state-preparation and measurement errors using single-qubit operations”

References

- [1] Quantum error correction below the surface code threshold. *Nature* 638, 8052 (2025), 920–926.
- [2] ANDRADE, B., DAVOUDI, Z., GRASS, T., HAFEZI, M., PAGANO, G., AND SEIF, A. Engineering an effective three-spin hamiltonian in trapped-ion systems for applications in quantum simulation. *Quantum Science and Technology* 7, 3 (apr 2022), 034001.
- [3] ANTHONY-PETERSEN, R., BIEKERT, A., BUNKER, R., CHANG, C. L., CHANG, Y.-Y., CHAPLINSKY, L., FASCIONE, E., FINK, C. W., GARCIA-SCIVERES, M., GERMOND, R., ET AL. A stress-induced source of phonon bursts and quasiparticle poisoning. *Nature Communications* 15, 1 (2024), 6444.
- [4] BAO, W., JIN, S., AND MARKOWICH, P. A. On time-splitting spectral approximations for the schrödinger equation in the semiclassical regime. *Journal of Computational Physics* 175, 2 (2002), 487–524.
- [5] BARENDTS, R., KELLY, J., MEGRANT, A., VEITIA, A., SANK, D., JEFFREY, E., WHITE, T. C., MUTUS, J., FOWLER, A. G., CAMPBELL, B., ET AL. Superconducting quantum circuits at the surface code threshold for fault tolerance. *Nature* 508, 7497 (2014), 500–503.
- [6] BAUM, Y., AMICO, M., HOWELL, S., HUSH, M., LIUZZI, M., MUNDADA, P., MERKH, T., CARVALHO, A. R., AND BIERCUK, M. J. Experimental deep reinforcement learning for error-robust gate-set design on a superconducting quantum computer. *PRX Quantum* 2, 4 (2021), 040324.

- [7] BAYRAKTAR, H., CHARARA, A., CLARK, D., COHEN, S., COSTA, T., FANG, Y.-L. L., GAO, Y., GUAN, J., GUNNELS, J., HAIDAR, A., HEHN, A., HOHNERBACH, M., JONES, M., LUBOWE, T., LYAKH, D., MORINO, S., SPRINGER, P., STANWYCK, S., TERENTYEV, I., VARADHAN, S., WONG, J., AND YAMAGUCHI, T. cuquantum sdk: A high-performance library for accelerating quantum science. In *2023 IEEE International Conference on Quantum Computing and Engineering (QCE)* (2023), vol. 01, pp. 1050–1061.
- [8] BENEDETTI, M., LLOYD, E., SACK, S., AND FIORENTINI, M. Parameterized quantum circuits as machine learning models. *Quantum Science and Technology* 4, 4 (nov 2019), 043001.
- [9] BIAMONTE, J., WITTEK, P., PANCOTTI, N., REBENTROST, P., WIEBE, N., AND LLOYD, S. Quantum machine learning. *Nature* 549, 7671 (2017), 195–202.
- [10] BLUVSTEIN, D., EVERED, S. J., GEIM, A. A., LI, S. H., ZHOU, H., MANOVITZ, T., EBADI, S., CAIN, M., KALINOWSKI, M., HANGLEITER, D., ET AL. Logical quantum processor based on reconfigurable atom arrays. *Nature* 626, 7997 (2024), 58–65.
- [11] BRAVO-PRIETO, C., LAROSE, R., CEREZO, M., SUBASI, Y., CINCIO, L., AND COLES, P. J. Variational quantum linear solver. *Quantum* 7 (2023), 1188.
- [12] BRAVYI, S. B., AND KITAEV, A. Y. Fermionic quantum computation. *Annals of Physics* 298, 1 (2002), 210–226.
- [13] CADI TAZI, L., AND THOM, A. J. Folded spectrum vqe: A quantum computing method for the calculation of molecular excited states. *Journal of Chemical Theory and Computation* 20, 6 (2024), 2491–2504.
- [14] CALLISON, A., AND CHANCELLOR, N. Hybrid quantum-classical algorithms in the noisy intermediate-scale quantum era and beyond. *Physical Review A* 106, 1 (2022), 010101.
- [15] CEREZO, M., ARRASMITH, A., BABBUSH, R., BENJAMIN, S. C., ENDO, S., FUJII, K., MCCLEAN, J. R., MITARAI, K., YUAN, X., CINCIO, L., ET AL. Variational quantum algorithms. *Nature Reviews Physics* 3, 9 (2021), 625–644.
- [16] CHENG, G., AND GUO, J. Modeling correlated-noise in silicon spin qubit device. *APL Quantum* 2, 1 (2025).
- [17] DALTON, K., LONG, C. K., YORDANOV, Y. S., SMITH, C. G., BARNES, C. H., MERTIG, N., AND ARVIDSSON-SHUKUR, D. R. Quantifying the effect of gate errors on variational quantum eigensolvers for quantum chemistry. *npj Quantum Information* 10, 1 (2024), 18.

- [18] DARIBAYEV, B., MUKHANBET, A., AND IMANKULOV, T. Implementation of the hhl algorithm for solving the poisson equation on quantum simulators. *Applied Sciences* 13, 20 (2023), 11491.
- [19] DENNIS, E., KITAEV, A., LANDAHL, A., AND PRESKILL, J. Topological quantum memory. *Journal of Mathematical Physics* 43, 9 (2002), 4452–4505.
- [20] DI BARTOLOMEO, G., VISCHI, M., CESA, F., WIXINGER, R., GROSSI, M., DONADI, S., AND BASSI, A. Noisy gates for simulating quantum computers. *Physical Review Research* 5, 4 (2023), 043210.
- [21] DING, Z., AND LIN, L. Simultaneous estimation of multiple eigenvalues with short-depth quantum circuit on early fault-tolerant quantum computers. *Quantum* 7 (2023), 1136.
- [22] EASTIN, B., AND KNILL, E. Restrictions on transversal encoded quantum gate sets. *Physical review letters* 102, 11 (2009), 110502.
- [23] EVERED, S. J., BLUVSTEIN, D., KALINOWSKI, M., EBADI, S., MANOVITZ, T., ZHOU, H., LI, S. H., GEIM, A. A., WANG, T. T., MASKARA, N., ET AL. High-fidelity parallel entangling gates on a neutral-atom quantum computer. *Nature* 622, 7982 (2023), 268–272.
- [24] FEI, X., BRADY, L., LARSON, J., LEYFFER, S., AND SHEN, S. Switching time optimization for binary quantum optimal control. *ACM Transactions on Quantum Computing* 6, 1 (2025), 1–38.
- [25] FEI, X., BRADY, L. T., LARSON, J., LEYFFER, S., AND SHEN, S. Binary control pulse optimization for quantum systems. *Quantum* 7 (2023), 892.
- [26] FEIT, M., FLECK, J., AND STEIGER, A. Solution of the schrödinger equation by a spectral method. *Journal of Computational Physics* 47, 3 (1982), 412–433.
- [27] FEYNMAN, R. P. Simulating physics with quantum computers. *International Journal of Theoretical Physics* 21, 7 (1982), 467–488.
- [28] FEYNMAN, R. P. Quantum mechanical computers. *Foundations of Physics* 16 (1986), 507–531. originally published in Optics News, February 1985, pp. 11-20.
- [29] GAMBLE, S. Quantum computing: What it is, why we want it, and how we’re trying to get it. In *Frontiers of Engineering: Reports on Leading-Edge Engineering from the 2018 Symposium* (2019), National Academies Press, pp. 5–8.
- [30] GILYÉN, A., SU, Y., LOW, G. H., AND WIEBE, N. Quantum singular value transformation and beyond: exponential improvements for quantum matrix arithmetics. In *Proceedings of the 51st annual ACM SIGACT symposium on theory of computing* (2019), pp. 193–204.

- [31] GONTHIER, J. F., RADIN, M. D., BUDA, C., DOSKOCIL, E. J., ABUAN, C. M., AND ROMERO, J. Measurements as a roadblock to near-term practical quantum advantage in chemistry: Resource analysis. *Physical Review Research* 4, 3 (2022), 033154.
- [32] GOPALAKRISHNAN MEENA, M., GOTTIPARTHI, K. C., LIETZ, J. G., GEORGIADOU, A., AND COELLO PÉREZ, E. A. Solving the hele–shaw flow using the harrow–hassidim–lloyd algorithm on superconducting devices: A study of efficiency and challenges. *Physics of Fluids* 36, 10 (2024).
- [33] GRIVOPOULOS, S. *Optimal control of quantum systems*. University of California, Santa Barbara, 2005.
- [34] GUERRESCHI, G. G., HOGABOAM, J., BARUFFA, F., AND SAWAYA, N. P. D. Intel Quantum Simulator: a cloud-ready high-performance simulator of quantum circuits. *Quantum Science and Technology* 5, 3 (July 2020).
- [35] GUNDLACH, H., SHARKEY, K., LYNCH, J., HAZOGLU, V., HSU, K.-C., DUKATZ, C., CRANE, E., WALCZYK, K., BODZIAK, M., GALATSANOS-DUECK, J., AND THOMPSON, N. Quantum advantage in computational chemistry?, 2025.
- [36] GÜNTHER, S., AND PETERSSON, N. A. A practical approach to determine minimal quantum gate durations using amplitude-bounded quantum controls. *AVS Quantum Science* 5, 4 (2023).
- [37] GÜNTHER, S., PETERSSON, N. A., AND DUBOIS, J. L. Quandary: An open-source c++ package for high-performance optimal control of open quantum systems. In *2021 IEEE/ACM Second International Workshop on Quantum Computing Software (QCS)* (2021), pp. 88–98.
- [38] HAN, J., ZHANG, L., AND E, W. Solving many-electron schrödinger equation using deep neural networks. *Journal of Computational Physics* 399 (2019), 108929.
- [39] HANGLEITER, D., ROTH, I., FUKSA, J., EISERT, J., AND ROUSHAN, P. Robustly learning the hamiltonian dynamics of a superconducting quantum processor. *Nature Communications* 15, 1 (2024), 9595.
- [40] HARROW, A. W., HASSIDIM, A., AND LLOYD, S. Quantum algorithm for linear systems of equations. *Physical review letters* 103, 15 (2009), 150502.
- [41] HE, J., AND WU, D. Zeroing neural dynamics solving time-variant complex conjugate matrix equation. *arXiv preprint arXiv:2406.12783* (2024).
- [42] HE, W., LI, T., LI, X., LI, Z., WANG, C., AND WANG, K. Efficient optimal control of open quantum systems, 2024.

- [43] HERMANN, J., SCHÄTZLE, Z., AND NOÉ, F. Deep-neural-network solution of the electronic schrödinger equation. *Nature Chemistry* 12, 10 (2020), 891–897.
- [44] HERRMANN, N., ARYA, D., DOHERTY, M. W., MINGARE, A., PILLAY, J. C., PREIS, F., AND PRESTEL, S. Quantum utility—definition and assessment of a practical quantum advantage. In *2023 IEEE International Conference on Quantum Software (QSW)* (2023), IEEE, pp. 162–174.
- [45] HIBAT-ALLAH, M., MAURI, M., CARRASQUILLA, J., AND PERDOMO-ORTIZ, A. A framework for demonstrating practical quantum advantage: comparing quantum against classical generative models. *Communications Physics* 7, 1 (2024), 68.
- [46] IDE, B., GOWDA, M. G., NADKARNI, P. J., AND DAUPHINAIS, G. Fault-tolerant logical measurements via homological measurement. *Physical Review X* 15, 2 (2025), 021088.
- [47] ITANI, W., SREENIVASAN, K. R., AND SUCCI, S. Quantum algorithm for lattice boltzmann (qalb) simulation of incompressible fluids with a nonlinear collision term. *Physics of Fluids* 36, 1 (2024).
- [48] JAVADI-ABHARI, A., TREINISH, M., KRSULICH, K., WOOD, C. J., LISHMAN, J., GACON, J., MARTIEL, S., NATION, P. D., BISHOP, L. S., CROSS, A. W., ET AL. Quantum computing with qiskit. *arXiv preprint arXiv:2405.08810* (2024).
- [49] JIN, S., LI, X., LIU, N., AND YU, Y. Quantum simulation for quantum dynamics with artificial boundary conditions. *SIAM Journal on Scientific Computing* 46, 4 (2024), B403–B421.
- [50] JIN, S., LIU, N., MA, C., AND YU, Y. Quantum preconditioning method for linear systems problems via schrödingerization. *arXiv preprint arXiv:2505.06866* (2025).
- [51] JIN, S., LIU, N., AND YU, Y. Quantum circuits for the heat equation with physical boundary conditions via schrodingerisation. *arXiv preprint arXiv:2407.15895* (2024).
- [52] JONES, T., AND BENJAMIN, S. Questlink—mathematica embiggened by a hardware-optimised quantum emulator*. *Quantum Science and Technology* 5, 3 (May 2020), 034012.
- [53] KANAI, T., JIN, D., AND GUO, W. Single-electron qubits based on quantum ring states on solid neon surface. *Phys. Rev. Lett.* 132 (Jun 2024), 250603.
- [54] KANG, J.-H., AND RYU, H. PennyLane-lightning mpi: A massively scalable quantum circuit simulator based on distributed computing in cpu clusters. *arXiv preprint arXiv:2508.13615* (2025).

- [55] KEIP, S., CAMPS, D., AND BEEUMEN, R. V. Qclab: A matlab toolbox for quantum computing, 2025.
- [56] KHAIT, I., THAM, E., SEGAL, D., AND BRODUTCH, A. Variational quantum eigensolvers in the era of distributed quantum computers. *Physical Review A* 108, 5 (2023), L050401.
- [57] KHAN, M. Q., NORRIS, L. M., AND VIOLA, L. Separate and efficient characterization of state-preparation and measurement errors using single-qubit operations. *arXiv preprint arXiv:2509.19448* (2025).
- [58] KIM, Y., EDDINS, A., ANAND, S., WEI, K. X., VAN DEN BERG, E., ROSENBLATT, S., NAYFEH, H., WU, Y., ZALETEL, M., TEMME, K., ET AL. Evidence for the utility of quantum computing before fault tolerance. *Nature* 618, 7965 (2023), 500–505.
- [59] KITAEV, A. Y. Quantum measurements and the abelian stabilizer problem. *arXiv preprint quant-ph/9511026* (1995).
- [60] KITAEV, A. Y. Fault-tolerant quantum computation by anyons. *Annals of physics* 303, 1 (2003), 2–30.
- [61] KNILL, E., AND LAFLAMME, R. Theory of quantum error-correcting codes. *Physical Review A* 55, 2 (1997), 900.
- [62] KO, T., LI, X., AND WANG, C. Implementation of the density-functional theory on quantum computers with linear scaling with respect to the number of atoms. *arXiv preprint arXiv:2307.07067* (2023).
- [63] KRINNER, S., LACROIX, N., REMM, A., DI PAOLO, A., GENOIS, E., LEROUX, C., HELLINGS, C., LAZAR, S., SWIADEK, F., HERRMANN, J., ET AL. Realizing repeated quantum error correction in a distance-three surface code. *Nature* 605, 7911 (2022), 669–674.
- [64] KUBISCHTA, E., AND TEIXEIRA, I. Family of quantum codes with exotic transversal gates. *Physical Review Letters* 131, 24 (2023), 240601.
- [65] LIAO, S., LIU, J., XIAO, X., FU, D., WANG, G., AND JIN, L. Modified gradient neural networks for solving the time-varying sylvester equation with adaptive coefficients and elimination of matrix inversion. *Neurocomputing* 379 (2020), 1–11.
- [66] LIDAR, D. A. Lecture notes on the theory of open quantum systems. *arXiv preprint arXiv:1902.00967* (2019).
- [67] LIN, L. Dissipative preparation of many-body quantum states: Toward practical quantum advantage. *APL Computational Physics* 1, 1 (2025), 010901.
- [68] LIN, L., SAAD, Y., AND YANG, C. Approximating spectral densities of large matrices. *SIAM review* 58, 1 (2016), 34–65.

- [69] LOTSHAW, P. C., BATTLES, K. D., GARD, B., BUCHS, G., HUMBLE, T. S., AND HEROLD, C. D. Modeling noise in global mølmer-sørensen interactions applied to quantum approximate optimization. *Phys. Rev. A* 107 (Jun 2023), 062406.
- [70] LOTSHAW, P. C., SAWYER, B. C., HEROLD, C. D., AND BUCHS, G. Exactly solvable model of light-scattering errors in quantum simulations with metastable trapped-ion qubits. *Phys. Rev. A* 110 (Sep 2024), L030803.
- [71] LOTSHAW, P. C., XU, H., KHALID, B., BUCHS, G., HUMBLE, T. S., AND BANERJEE, A. Simulations of frustrated ising hamiltonians using quantum approximate optimization. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 381, 2241 (2023), 20210414.
- [72] LUO, D., SURACE, F. M., DE, A., LEROSE, A., BENNEWITZ, E. R., WARE, B., SCHUCKERT, A., DAVOUDI, Z., GORSHKOV, A. V., KATZ, O., AND MONROE, C. Quantum simulation of bubble nucleation across a quantum phase transition, 2025.
- [73] MACIEJEWSKI, F. B., BIAMONTE, J., HADFIELD, S., AND VENTURELLI, D. Improving quantum approximate optimization by noise-directed adaptive remapping, 2024.
- [74] MACK, C. The multiple lives of moore’s law. *IEEE Spectrum* 52, 4 (2015), 31–31.
- [75] MARKOV, I. L. Limits on fundamental limits to computation. *Nature* 512, 7513 (2014), 147–154.
- [76] MARRA, P. Majorana nanowires for topological quantum computation. *Journal of Applied Physics* 132, 23 (2022).
- [77] MOORE, G. Cramming more components onto integrated circuits (1965).
- [78] NAUTRUP, H. P., DELFOSSE, N., DUNJKO, V., BRIEGEL, H. J., AND FRIIS, N. Optimizing quantum error correction codes with reinforcement learning. *Quantum* 3 (2019), 215.
- [79] NGHIEM, N. A., AND WEI, T.-C. Quantum algorithm for estimating largest eigenvalues. *Physics Letters A* 488 (2023), 129138.
- [80] NGUYEN, T., LYAKH, D., DUMITRESCU, E., CLARK, D., LARKIN, J., AND MCCASKEY, A. Tensor network quantum virtual machine for simulating quantum circuits at exascale. *ACM Transactions on Quantum Computing* 4, 1 (2022), 1–21.
- [81] NIELSEN, M. A., AND CHUANG, I. L. *Quantum computation and quantum information*. Cambridge university press, 2010.

- [82] NÜSSELER, A., DHAND, I., HUELGA, S. F., AND PLENIO, M. B. Efficient simulation of open quantum systems coupled to a fermionic bath. *Phys. Rev. B* 101 (Apr 2020), 155134.
- [83] PUTTERMAN, H., NOH, K., HANN, C. T., MACCABE, G. S., AGHAEIMEIBODI, S., PATEL, R. N., LEE, M., JONES, W. M., MORADINEJAD, H., RODRIGUEZ, R., MAHULI, N., ROSE, J., OWENS, J. C., LEVINE, H., ROSENFELD, E., REINHOLD, P., MONCELSI, L., ALCID, J. A., ALIDOUST, N., ARRANGOIZ-ARRIOLA, P., BARNETT, J., BIENIAS, P., CARSON, H. A., CHEN, C., CHEN, L., CHINKEZIAN, H., CHISHOLM, E. M., CHOU, M.-H., CLERK, A., CLIFFORD, A., COSMIC, R., CURIEL, A. V., DAVIS, E., DELORENZO, L., D’EWART, J. M., DIKY, A., D’SOUZA, N., DUMITRESCU, P. T., EISENMANN, S., ELKHOULY, E., EVENBLY, G., FANG, M. T., FANG, Y., FLING, M. J., FON, W., GARCIA, G., GORSHKOV, A. V., GRANT, J. A., GRAY, M. J., GRIMBERG, S., GRIMSMO, A. L., HAIM, A., HAND, J., HE, Y., HERNANDEZ, M., HOVER, D., HUNG, J. S. C., HUNT, M., IVERSON, J., JARRIGE, I., JASKULA, J.-C., JIANG, L., KALAEI, M., KARABALIN, R., KARALEKAS, P. J., KELLER, A. J., KHALAJHEDAYATI, A., KUBICA, A., LEE, H., LEROUX, C., LIEU, S., LY, V., MADRIGAL, K. V., MARCAUD, G., MCCABE, G., MILES, C., MILSTED, A., MINGUZZI, J., MISHRA, A., MUKHERJEE, B., NAGHILOO, M., OBLEPIAS, E., ORTUNO, G., PAGDILAO, J., PANCOTTI, N., PANDURO, A., PAQUETTE, J., PARK, M., PEAIRS, G. A., PERELLO, D., PETERSON, E. C., PONTE, S., PRESKILL, J., QIAO, J., REFAEL, G., RESNICK, R., RETZKER, A., REYNA, O. A., RUNYAN, M., RYAN, C. A., SAHMOUD, A., SANCHEZ, E., SANIL, R., SANKAR, K., SATO, Y., SCAFFIDI, T., SIAVOSHI, S., SIVARAJAH, P., SKOGLAND, T., SU, C.-J., SWENSON, L. J., TEO, S. M., TOMADA, A., TORLAI, G., WOLLACK, E. A., YE, Y., ZERRUDO, J. A., ZHANG, K., BRANDÃO, F. G. S. L., MATHENY, M. H., AND PAINTER, O. Hardware-efficient quantum error correction via concatenated bosonic qubits. *Nature* 638, 8052 (Feb. 2025), 927–934.
- [84] QING, M., AND XIE, W. Use vqe to calculate the ground energy of hydrogen molecules on ibm quantum, 2023.
- [85] RAMKARTHIK, M., AND SOLANKI, P. D. *Numerical recipes in quantum information theory and quantum computing: an adventure in FORTRAN 90*. CRC Press, 2021.
- [86] ROFFE, J. Quantum error correction: an introductory guide. *Contemporary Physics* 60, 3 (2019), 226–245.
- [87] RYU, J., JUNG, J., LEE, J., GWON, Y., AND KEVIN RHEE, J. A variational quantum algorithm for ordered svd. In *2021 IEEE Global Communications Conference (GLOBECOM)* (2021), pp. 01–05.

- [88] SAWAYA, N. P., MARTI-DAFCIK, D., HO, Y., TABOR, D. P., BERNAL NEIRA, D. E., MAGANN, A. B., PREMARATNE, S., DUBEY, P., MATSUURA, A., BISHOP, N., DE JONG, W. A., BENJAMIN, S., PAREKH, O. D., TUBMAN, N. M., KLYMKO, K., AND CAMPS, D. Hamlib: A library of hamiltonians for benchmarking quantum algorithms and hardware. In *2023 IEEE International Conference on Quantum Computing and Engineering (QCE)* (2023), vol. 02, pp. 389–390.
- [89] SELSTØ, S., AND KVAAL, S. Absorbing boundary conditions for dynamical many-body quantum systems. *Journal of Physics B: Atomic, Molecular and Optical Physics* 43, 6 (mar 2010), 065004.
- [90] SHALF, J. The future of computing beyond moore’s law. *Philosophical Transactions of the Royal Society A* 378, 2166 (2020), 20190061.
- [91] SHIRAKAWA, T., UEDA, H., AND YUNOKI, S. Automatic quantum circuit encoding of a given arbitrary quantum state. *Phys. Rev. Res.* 6 (Oct 2024), 043008.
- [92] SHOR, P. W. Scheme for reducing decoherence in quantum computer memory. *Physical review A* 52, 4 (1995), R2493.
- [93] SHOR, P. W. Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer. *SIAM Review* 41, 2 (1999), 303–332.
- [94] SHUKRINOV, Y. M., RAHMONOV, I., KULIKOV, K., BOTHA, A., PLECENIK, A., SEIDEL, P., AND NAWROCKI, W. Modeling of lc-shunted intrinsic josephson junctions in high-tc superconductors. *Superconductor Science and Technology* 30, 2 (2016), 024006.
- [95] STETINA, T., AND WIEBE, N. First-quantized quantum simulation of non-relativistic qed with emergent topologically protected coulomb interactions, 2025.
- [96] SUN, H., AND GALPERIN, M. Control of open quantum systems: The nonequilibrium green’s function perspective. *APL Quantum* 2, 1 (2025).
- [97] TEAM, Q. A., AND COLLABORATORS. qsim, Sept. 2020.
- [98] TILLY, J., CHEN, H., CAO, S., PICOZZI, D., SETIA, K., LI, Y., GRANT, E., WOSSNIG, L., RUNGGER, I., BOOTH, G. H., ET AL. The variational quantum eigensolver: a review of methods and best practices. *Physics Reports* 986 (2022), 1–128.
- [99] VIAMONTES, G. F., MARKOV, I. L., AND HAYES, J. P. High-performance quidd-based simulation of quantum circuits. In *Proceedings Design, Automation and Test in Europe Conference and Exhibition* (2004), vol. 2, IEEE, pp. 1354–1355.

- [100] WECKER, D., AND SVORE, K. M. LIQUi—j: A Software Design Architecture and Domain-Specific Language for Quantum Computing, 2014.
- [101] WILLE, R., FOWLER, A., AND NAVEH, Y. Computer-aided design for quantum computation. In *2018 IEEE/ACM International Conference on Computer-Aided Design (ICCAD)* (2018), IEEE, pp. 1–6.
- [102] WINTERSPERGER, K., DOMMERT, F., EHMER, T., HOURSANOV, A., KLEPSCH, J., MAUERER, W., REUBER, G., STROHM, T., YIN, M., AND LUBER, S. Neutral atom quantum computing hardware: performance and end-user perspective. *EPJ Quantum Technology* 10, 1 (2023), 32.
- [103] ZHANG, J., PAGANO, G., HESS, P. W., KYPRIANIDIS, A., BECKER, P., KAPLAN, H., GORSHKOV, A. V., GONG, Z.-X., AND MONROE, C. Observation of a many-body dynamical phase transition with a 53-qubit quantum simulator. *Nature* 551, 7682 (2017), 601–604.
- [104] ZHANG, Q., SALIGANE, M., KIM, H.-S., BLAAUW, D., TZIMPRAGOS, G., AND SYLVESTER, D. Quantum circuit simulation with fast tensor decision diagram. In *2024 25th International Symposium on Quality Electronic Design (ISQED)* (2024), IEEE, pp. 1–8.
- [105] ZHENG, W., YING, L.-A., AND DING, P. Numerical solutions of the schrödinger equation for the ground lithium by the finite element method. *Applied mathematics and computation* 153, 3 (2004), 685–695.
- [106] ZOU, X., AND LO, H.-K. Algebraic topology principles behind topological quantum error correction. *arXiv preprint arXiv:2505.06082* (2025).
- [107] ZOUFAL, C., LUCCHI, A., AND WOERNER, S. Quantum generative adversarial networks for learning and loading random distributions. *npj Quantum Information* 5, 1 (2019), 103.